

Problem Set 1

September 23, 2026

(solutions are due on September 30, 2026, 8h30)

Problem 1 Conditional Mutual Information and Markov Chains

Let X, Y, Z be discrete random variables.

1. Show that

$$I(X; Y | Z) = 0$$

if and only if $X - Z - Y$ form a Markov chain, i.e., if and only if

$$P_{XY|Z}(x, y | z) = P_{X|Z}(x | z)P_{Y|Z}(y | z)$$

for every z with $P_Z(z) > 0$.

2. Construct an example of three binary random variables X, Y, Z for which

$$I(X; Y) = 0 \quad \text{but} \quad I(X; Y | Z) > 0.$$

Compute both quantities explicitly and explain why $X - Z - Y$ does not form a Markov chain.

Problem 2

Mixing Increases Entropy

1. Let

$$p = (p_1, \dots, p_j, \dots, p_k, \dots, p_m)$$

be a probability distribution, and fix $\alpha \in [0, 1]$. Define q by leaving all entries except j and k unchanged and setting

$$q_j = \alpha p_j + (1 - \alpha) p_k, \quad q_k = (1 - \alpha) p_j + \alpha p_k.$$

Show that $H(q) \geq H(p)$. When $\alpha = \frac{1}{2}$, interpret the operation in words.

2. Let $q = (q_1, \dots, q_m)$ be a probability distribution for which the endpoint masses $q_1 = a$ and $q_m = b$ are fixed, where $a, b \geq 0$ and $a + b \leq 1$. Find the smallest and largest possible values of $H(q)$. Specify probability distributions that attain the two bounds.

Problem 3

Entropy of Functions of a Random Variable

Let X be a discrete random variable taking values in a finite alphabet $\mathcal{X}$, and let $U = f(X)$ for a deterministic function f . Justify the following steps:

$$H(X, U) \stackrel{(a)}{=} H(X) + H(U | X) \stackrel{(b)}{=} H(X),$$

$$H(X, U) \stackrel{(c)}{=} H(U) + H(X | U) \stackrel{(d)}{\geq} H(U).$$

Conclude that $H(f(X)) \leq H(X)$.

- Explain briefly why this inequality is intuitively reasonable.
- Characterize precisely when equality $H(f(X)) = H(X)$ holds. Your statement should take into account that some symbols in $\mathcal{X}$ may have probability zero.

Problem 4 Mutual Information and Conditional Mutual Information

Consider a sequence of n (dependent) binary random variables $X_1, \dots, X_n$. The joint law $P_{X_1, \dots, X_n}$ is described as follows. For every sequence $x_1, \dots, x_n$ that has an *even* number of 1's,

$$P_{X_1, \dots, X_n}(x_1, \dots, x_n) = 2^{-(n-1)}.$$

For every sequence $x_1, \dots, x_n$ that has an *odd* number of 1's,

$$P_{X_1, \dots, X_n}(x_1, \dots, x_n) = 0.$$

1. Compute $I(X_1; X_2)$ when $n = 10$ and when $n = 2$.
2. Compute $I(X_1; X_2 \mid X_3)$ for $n \geq 3$. How does it depend on n ?
3. Compute $I(X_1, \dots, X_{n-1}; X_n)$ and $I(X_1; X_n \mid X_2, \dots, X_{n-1})$.